SALTDRAG · PRECIS

SaltDrag design precis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SDG-PRC-001	Version	0.4
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/02-concept.md @ c7f0938		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-24	Amish Chadha	Kit 1.8 and 1.9.1 upgrade; numbers from SDG-CAL-001 v0.3 after the STEM check fixes (SDG-DEC-001 D17 to D24); concept options and STEM check referenced; one set per family pan
0.3	2026-10-03	Amish Chadha	TRL 3; numbers from SDG-CAL-001 and the constructable design (SDG-DDR-002); bund roller and ballast added, windrow board removed
0.2	2026-10-03	Amish Chadha	TRL 2 precis; how it works, components, key design choices, first-order numbers, safety and concept media (SDG-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SaltDrag design precis

Drags a rake across the salt pan with hand capstans on the bunds so workers stay out of the brine.

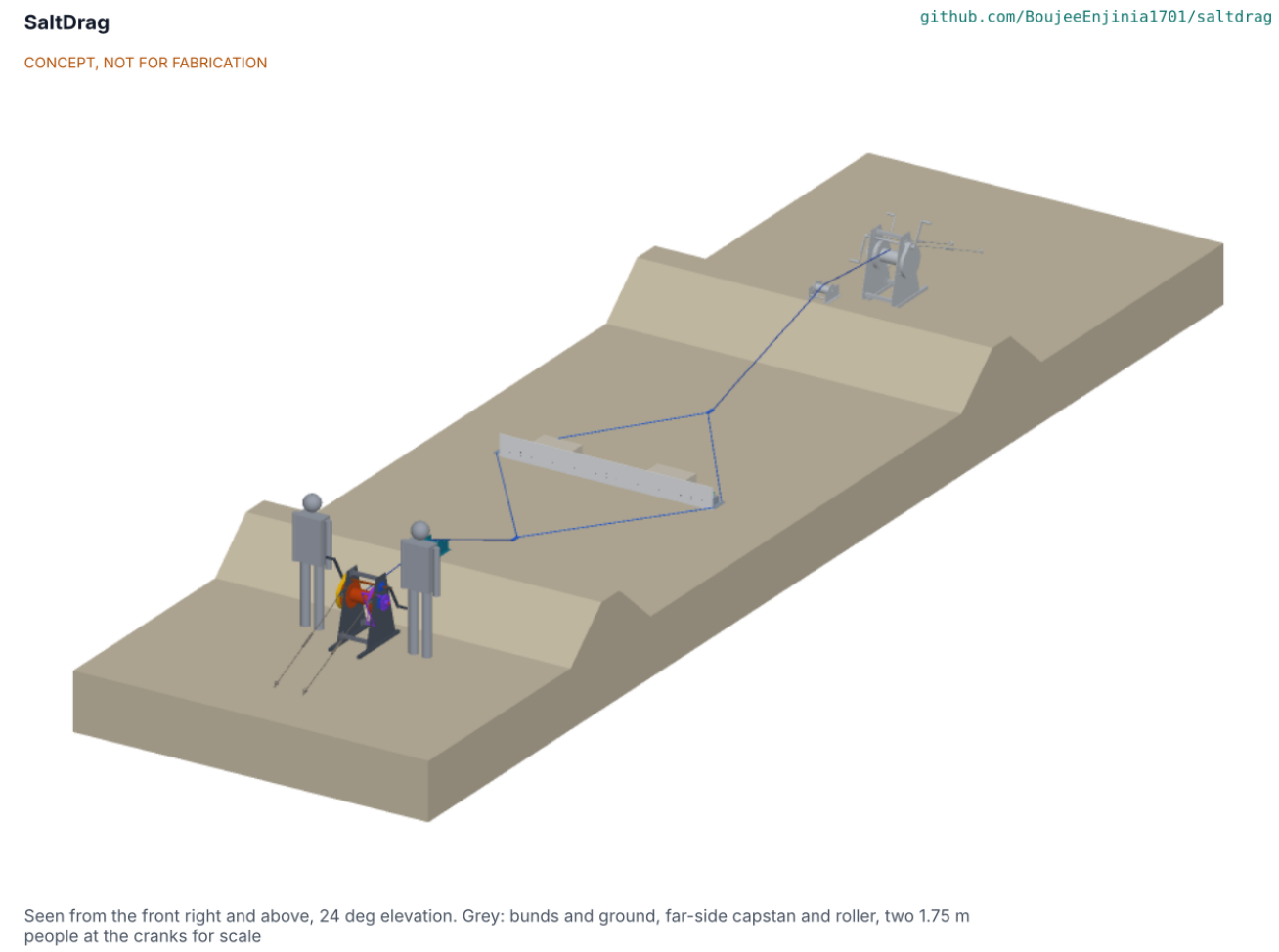


Figure 1. The working layout on a pan shortened to 7 m for the picture: the near capstan and bund roller in colour, the far-side set in grey, two 1.75 m people at the cranks for scale (concept media, SDG-DWG-010).

How it works

Two people winding a hand capstan on dry ground behind one bund draw a 3 m rake across the pan by a rope that runs over a roller on the bund crest. The rake's plastic blade rides 10 mm above the floor on three skids and pushes the loose salt crystals ahead of it into a heap of about 230 kg. At the near bund the steepening rope starts to lift the rake about 3.5 m from the bund toe, so the heap is left there as a windrow. A second capstan on the far bund holds a light back tension while the rake is pulled in, then winds it back empty. Both capstans then move 3 m along their bunds, and the return pull brings the rake onto the next strip. Nobody stands in the brine or walks on the pan at any stage.

Each capstan is a geared hand winch: two cranks drive a 14-tooth pinion, which turns an 84-tooth gear on the drum shaft (6 to 1). A pawl on a ratchet wheel stops the drum running back whenever the cranks are let go, and a band brake lets the far capstan pay out rope under control. Two chains from ears on the side frames, at rope height, hold the capstan to two screw-in ground anchors behind it.

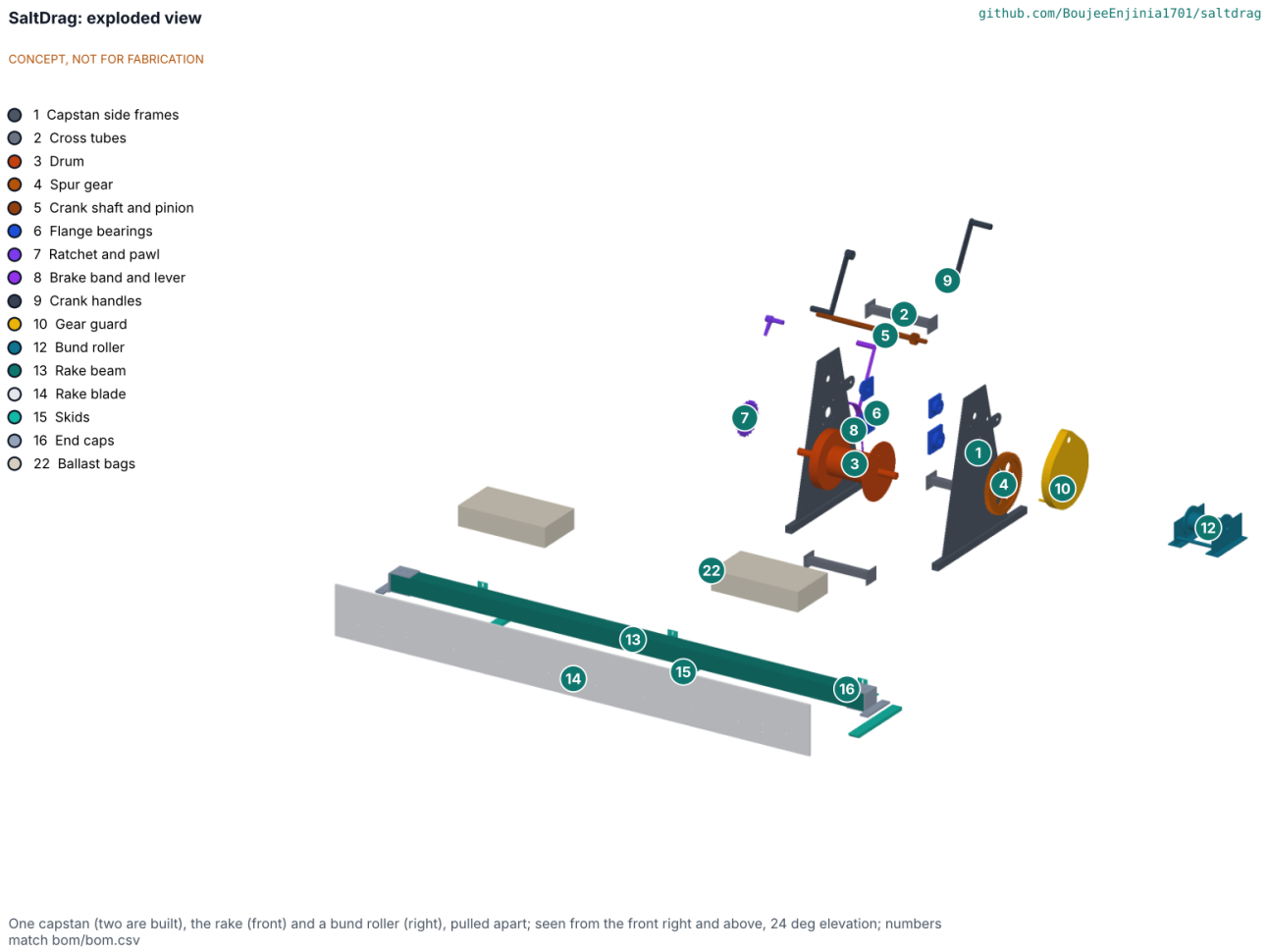


Figure 2. One capstan, the rake and a bund roller pulled apart; numbers match bom/bom.csv.



Figure 3. Capstan with its ratchet and brake side removed, showing the drum, flanges and three layers of rope.

Components

Table 1. Main components (two capstans, two rollers and two ropes in a set).

BOM	Component	Role
1, 2	Side frames and cross tubes	Two 6 mm galvanised cheeks on box-section runners, bolted together by three cross tubes with 100 x 80 mm end plates; the runners let the capstan be slid along the bund
3	Drum	168.3 x 6.3 mm YSt 310 barrel between 330 mm flanges on a 40 mm C45+QT shaft; holds 58 m of 12 mm rope; the left flange carries the brake rim
4, 5, 9	Gear, crank shaft and pinion, crank handles	Module 4 spur gears, 6 to 1, the gear on a 44 mm hub and the pinion teeth hardened; a stepped 30 and 25 mm crank shaft; two removable 330 mm cranks set 180 degrees apart
6	Flange bearings	Four-bolt bearings (40 mm and 30 mm bore) with stainless inserts in thermoplastic housings
7, 8	Ratchet and pawl; brake band and lever	The pawl holds the load on a ratchet with a welded hub; the band brake, with a 600 mm lever, controls paying out
10	Gear guard	Yellow sheet-steel cup over the gear mesh
11	Ground anchors, chains and turnbuckles	Two screw-in anchors per capstan, chained to the cheek ears with 10 mm chain and rated turnbuckles
12	Bund roller saddle	UHMW-PE roller on a small galvanised saddle astride the crest, so the rope never cuts the bund
13 to 16	Rake beam, blade, skids, end caps	3.0 m FRP box beam with a 300 mm UHMW-PE blade; skids set the blade height; stainless end caps carry the bridle eyes low, at the foot of the beam
17, 18	Bridles and pull ropes	Two-leg polyester bridles to a swivel; 45 m of 12 mm polyester double braid per side
20, 22	Rope dampers and ballast bags	A weighted bag on each loaded rope; two salt-filled bags keep the blade down near the bund
21	Maintenance kit	Rinse water, grease and spares

Key design choices

All were decided on 2026-10-03 under Amish's pre-approval of this batch (SDG-DDR-001 and SDG-DDR-002; register SDG-DEC-001). The kit 1.8 STEM check and its independent review found nine details that failed; each was fixed under Amish's standing instruction of 2026-10-23 without changing the working principle (SDG-STM-001; SDG-DEC-001 D17 to D24). Two other working principles, a bund-walk haul and a lever rope puller, were developed as concept options B and C; the three score within 10 % of each other, so the choice is Amish's (SDG-OPT-001, SDG-DDR-003). This precis describes option A, the design of record until he decides.

- **Drum winch, not a warping capstan.** The drum stores the rope, so nobody has to tail a loaded rope by hand.
- **Capstan behind the bund, roller on the crest.** A 0.4 m crest is too narrow for a capstan and two people cranking, and the roller keeps the rope off the bund.
- **Hand power, single spur stage.** No engine or motor, in keeping with the Agariya registration conditions.
- **Plastic and FRP in the pan, galvanised steel on dry ground.** Nothing that rusts sits in the brine.
- **One-piece 3 m rake with ballast.** The rake width keeps the design drag at the rated 3 kN; salt-filled bags keep the blade down as the rope steepens near the bund.

First-order numbers

Table 2. Key figures (estimates from SDG-CAL-001).

Quantity	Value	Basis
Full heap in front of the blade	0.19 m³, about 231 kg	300 mm blade, 35 degree repose, 1,200 kg/m³ loose salt
Design drag on the rake	2.75 kN	Heap sliding 1.59 kN, layer loosening 0.90 kN (assumed 300 N/m), skids 0.26 kN
Rope pull at the drum	3.0 kN	Drag, roller loss and 200 N back tension
Crank force, two people	89 N each	6 to 1, 330 mm cranks, 94 % gear and bearing efficiency, rope on its third layer; at the assumed drag
Pull with two people at 120 N	4.0 kN (3.7 kN on a full drum)	R3 asks for 3 kN
Rope speed and work rate	2.4 m/min at 23 crank turns a minute; 64 W each on average, 71 W at the end of a pass	Within the 75 W an adult sustains; early hours in the hot season
One strip	26 min: pull 12.4, return 5.7, move 8	30 m pan
Gathering rate	About 530 kg/h, 133 kg per worker-hour (crew of four); about 195 t in a 61-day harvest, a third of one pan's yearly output	Baseline for PR-005 still to be measured; one set per family pan
Heaviest single part	23.8 kg (ballast bag, filled on site); drum 23.5 kg	Model masses
Parts cost of the set	USD 2,897	Value-engineering target USD 4,000 (USD 1,103 under)

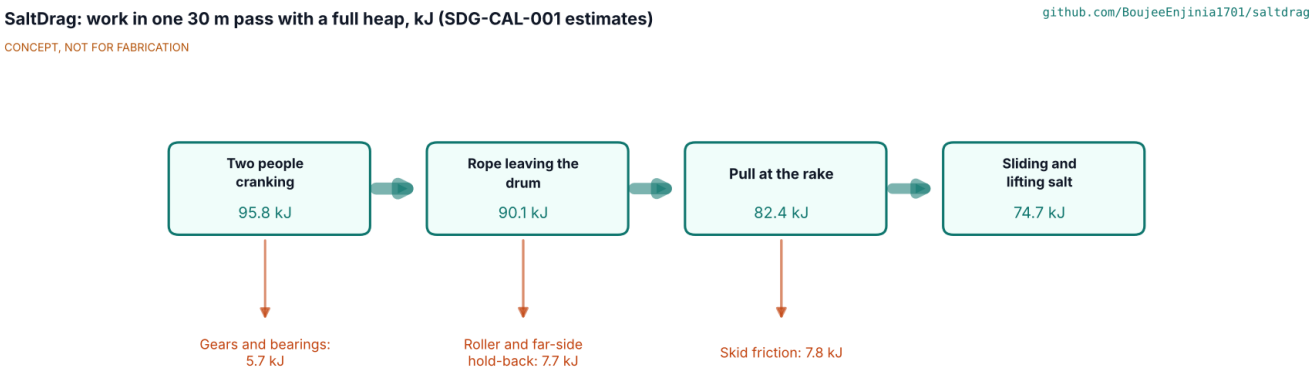


Figure 4. Where the cranking work goes in one 30 m pass with a full heap (estimates).

Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- No motorised or track-mounted auger harvester.
- Build on expired prior art such as the CA1243940A salt harvester; review CN102476810A (salt-pan rake) claims before release.

Shared blocks

- Hand capstan common block, shared with SiltHaul: this capstan is the first candidate for the common block (SDG-DDR-001). SiltHaul is not changed by this work.
- CalRig proof-load for rope, capstan and anchor load tests.

Safety

> **Safety:** A loaded rope stores energy and can whip back if it breaks, slips off the drum or loses its anchor. The rope is low-stretch polyester with a working-load factor of 7.7 at the rated pull, each loaded rope carries a weighted damper bag at mid-span, and nobody stands in line with a loaded rope or on the bund within 2 m of it.
>
> **Safety:** Cranks and gears can strike or trap hands. The pawl is always engaged while winding, so a released crank cannot spin back; the gears run inside a yellow guard; the crank handles are taken off before any rope is paid out under load, and paying out is done only on the brake.
>
> **Safety:** A ground anchor that pulls out lets the capstan lurch toward the pan. Each anchor is proof-pulled on site to 3.42 kN (1.5 times a 70 % share of the rated pull) before first use, and the rated 10 mm chains are fixed at rope height and kept equally tight so the capstan cannot tip.
>
> **Safety:** Heat is a serious hazard in the harvest season; plan work for cooler hours and provide shade and water. Parts of up to 24 kg are lifted and carried; lift in pairs.

This design is published as an open engineering reference, never as certified lifting or rope equipment. It is not certified equipment.

Open questions

No design decisions are open (register SDG-DEC-001). Facts that only a site survey or real parts can settle, such as the drag per metre on a real crystal bed, the bund crest width and the soil behind the bunds, are listed in the register to be confirmed.